

AD5941-Enabled Digital Active Electrode for Electrical Impedance Tomography

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Abstract—Electrical impedance tomography (EIT) is a functional imaging modality capable of reconstructing the conductivity distribution within the human body, which is challenged by limitations such as parasitic effects of long lead wires and system complexity. To address these issues, this paper proposes a novel digital active electrode architecture based on the AD5941, which mitigates signal degradation and simplifies system design by integrating all the analog front-end circuitry, including both excitation and measurement units, at the electrode site. An adaptive gain control mechanism is also introduced to improve the system's noise suppression capability and inter-channel consistency. A 16-electrode EIT system based on the proposed DAE has been developed and tested. Experiments demonstrate that the system achieves a signal-to-noise ratio over 70 dB at 62.5 kHz with 21.33 frames per second and is capable of reconstructing the positions and shapes of embedded objects in the saline tank.

Keywords—Electrical impedance tomography; digital active electrode; adaptive gain control; image reconstruction

I. INTRODUCTION

Electrical impedance tomography (EIT) is a promising functional imaging modality that can visualize the internal conductivity distribution of the human body, thereby affording the ability to detect functional or structural anomalies. With advantages of non-invasiveness, cost-effectiveness, and high temporal resolution, EIT has been extensively investigated for diverse clinical scenarios, including lung ventilation monitoring and brain function assessment [1], [2].

EIT works by first attaching multiple electrodes to the human body surface over the region of interest. Small alternating currents (AC) are then injected into the body via a selected pair of electrodes. Simultaneously, the induced voltages on the boundary are measured across all the remaining electrodes, which are then used to reconstruct a visualization of the internal conductivity distribution. Accordingly, an EIT hardware system comprises three essential components: an excitation module, a measurement module, and a control unit. The excitation module is responsible for generating precise AC signals and selectively injecting them through predetermined pairs

of electrodes, following a predefined excitation pattern. The purpose of the measurement module is to acquire boundary voltages from all non-injecting electrodes and ensure high-fidelity capture of amplitude and phase information referenced to the excitation signal. The control unit's role is to ensure precise temporal synchronization between excitation and measurement under different excitation-measurement patterns.

In some of the EIT systems, the analog front-end (AFE) circuits of both the excitation and measurement modules are positioned away from the electrode array and connected to the electrodes via long lead wires [3], [4]. The stray capacitance of the long lead wires and the skin-electrode impedance form a low-pass filter. Variations in the skin-electrode impedance alter the filter response, causing voltage amplitude fluctuations and reduced acquisition stability. To address these challenges, active electrode-based EIT (AE-EIT) systems have been introduced, which integrate signal conditioning circuits directly at the electrode site to process electrical signals before driving the long lead wires, thereby suppressing signal fluctuation and improving signal-to-noise ratio (SNR) of the system [5], [6]. Taking active electrodes one step further, digital active electrode-based EIT (DAE-EIT) systems integrate both the excitation module and measurement module directly at the electrode site, enabling local digital demodulation [7], [8]. This architecture not only eliminates analog signal transmission through long lead wires but also reduces data bandwidth requirements, especially critical in high-channel-count systems.

Building on the DAE-EIT architecture described above, we propose an AD5941-enabled digital active electrode (DAE) for EIT. In contrast to prior DAE implementations, our DAE employs an analog differential acquisition architecture to enhance the common-mode rejection ratio. Additionally, we introduce an adaptive gain control mechanism that dynamically adjusts the gain of the measurement module for each DAE based on the amplitude of the acquired boundary voltage signal. This ensures optimal signal conditioning across DAEs at different

spatial positions and substantially improves the SNR of the system.

II. PROPOSED DIGITAL ACTIVE ELECTRODE ARCHITECTURE

A. DAE Architecture

The architecture of the proposed AD5941-enabled DAE is illustrated in Fig. 1. Each DAE is identical and connects to its adjacent DAE via a short flexible printed circuit (FPC) cable. It has five parts: (1) Measurement module. This module comprises a discrete programmable gain instrumentation amplifier (PGIA) and the AD5941 internal common measurement channel, used to acquire the voltage difference between adjacent electrode pairs. (2) Excitation module. This module generates constant-amplitude sinusoidal current excitation using the AD5941's internal direct digital synthesizer (DDS) and high-speed digital-to-analog converter (DAC), followed by an external voltage-controlled current source (VCCS). (3) AD5941. The AD5941 is a high-precision impedance measurement integrated circuit from Analog Devices, integrating a 1.6 MSPS 16-bit analog-to-digital converter (ADC), high-speed DAC, DDS, programmable gain amplifier (PGA), Discrete Fourier Transform (DFT) engine, and other AFE circuits and digital signal processing blocks. (4) Microcontroller Unit (MCU). The STM32G0 microcontroller is employed to manage timing control, parameter configuration, and communication protocol conversion, thereby reducing the number of external signal lines through local handling of control and data flows within each DAE. (5) Electrode connector. A simple ring-shaped connector for direct connection to a snap-in electrocardiogram or other types of electrode, ensuring quick setup and stable signal transmission.

B. Measurement Module

The measurement module consists of three building blocks: a double-pole/double-throw (DPDT) switch, a discrete PGIA, and the AD5941 internal common measurement channel.

For DAEs used to measure boundary voltages, their DPDT switches (denoted as MUX1) connect to two adjacent electrodes to acquire inter-electrode potential differences for EIT image reconstruction. In contrast, for DAEs responsible for current injection, MUX1 connects across the sampling resistor R_S in the excitation module, allowing real-time monitoring of the injected current.

Fig. 2 shows the structure of the discrete PGIA circuit. The discrete PGIA is built upon three operational amplifier (op-amp) instrumentation amplifier (IA) structures and can be divided into two stages. The voltage signal from the MUX1 is applied differentially to U_+ and U_- and then processed by the first stage of the three op-amp IA, where it is buffered and amplified. A dual-channel 4-to-1 analog switch (denoted as MUX4) selects among different feedback resistor combinations to achieve four discrete gain settings of $1\times$, $4.02\times$, $8.03\times$, and $12.06\times$, controlled via the A0 and A1 pins. The resistor network is implemented with $R_1 = R'_1 = 4.53\text{ k}\Omega$,

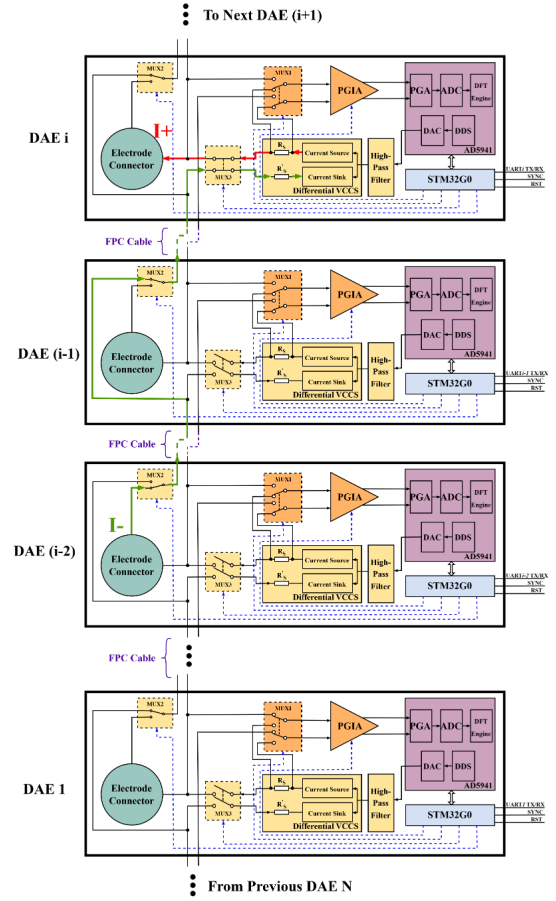


Fig. 1. Proposed DAE Architecture with N DAEs and a “skip-1” Excitation Pattern.

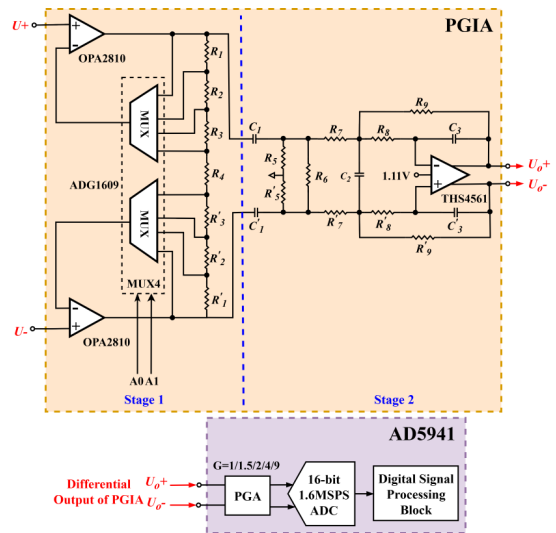


Fig. 2. Schematic of the PGIA and AD5941 common measurement channel.

$R_2 = R'_2 = 750 \Omega$, $R_3 = R'_3 = 250 \Omega$, and $R_4 = 1 \text{ k}\Omega$. The gain G of the differential amplifier is calculated by:

$$G = \frac{2R_F + R_G}{R_G} \quad (1)$$

where R_F is the selected feedback resistor, and R_G is the resistor that sets the gain. The specific resistor combinations and the corresponding gain values are summarized in TABLE I. The amplified signal from the first stage is subsequently

TABLE I
RESISTOR COMBINATIONS AND CORRESPONDING GAINS OF STAGE 1

G	$2R_F + R_G$	R_G
1	$2(R_1 + R_2 + R_3) + R_4$	$2(R_1 + R_2 + R_3) + R_4$
4.02	$2(R_1 + R_2 + R_3) + R_4$	$2(R_2 + R_3) + R_4$
8.03	$2(R_1 + R_2 + R_3) + R_4$	$2R_3 + R_4$
12.06	$2(R_1 + R_2 + R_3) + R_4$	R_4

conditioned by the second stage, which functions as a band-pass filter with a passband from 100 Hz to 600 kHz. This stage comprises a passive high-pass filter followed by a second-order Butterworth low-pass filter implemented using a THS4561 fully differential amplifier. The high-pass section removes DC offset, while the low-pass section attenuates high-frequency noise. The overall band-pass response provides unity gain, preserving signal amplitude while effectively attenuating out-of-band components prior to input to the AD5941 internal common measurement channel.

As shown in Fig. 2, the AD5941 internal common measurement channel integrates a PGA with five selectable gain settings ($1\times$, $1.5\times$, $2\times$, $4\times$, and $9\times$) at the front end. In conjunction with the discrete PGIA, the overall gain of the measurement module can be configured from 1 ($1\times$) up to 108.54 (12.06×9). After signal conditioning, the analog signal is digitized by the integrated ADC and demodulated by the on-chip DFT engine to extract the amplitude and phase of the measured voltage signal referenced to the excitation signal.

C. Excitation Module

The excitation module plays a crucial role in EIT systems, as accurate conductivity distribution reconstruction depends on the injected current remaining stable under different loads and frequencies. A differential VCCS based on Howland current source is implemented in the excitation module to convert the sinusoidal voltage generated by the AD5941 internal DDS and DAC into a constant-amplitude sinusoidal current signal [9], as shown in Fig. 3.

The AD5941 can produce sinusoidal voltage signals up to 200 kHz with an adjustable amplitude up to 750 mV, superimposed on a 1.11 V direct current (DC) bias voltage. A high-pass filter is employed to block DC before the signal drives the VCCS, preventing any DC from flowing through the human body. A differential VCCS based on the Howland current source is constructed using OPA2156 and OPA810, which reduces the load common-mode voltage and enhances the output swing compared with a single-ended VCCS. When

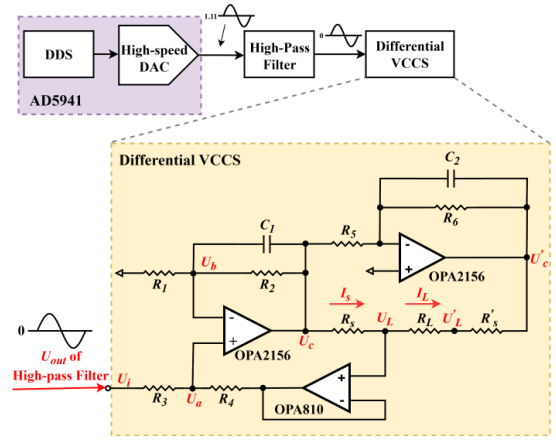


Fig. 3. Schematic of differential VCCS based on Howland current source.

$R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R$, the five key node voltages in the circuit are:

$$\begin{cases} U_a = \frac{R_3 \times (U_L - U_i)}{R_3 + R_4} + U_i = \frac{(U_L - U_i)}{2} + U_i \\ U_b = \frac{R_1 U_c}{R_1 + R_2} = \frac{U_c}{2} \\ U_L = \frac{(R_L + R'_s) \times (U_c - U'_c)}{R_L + R_s + R'_s} + U'_c \\ U'_c = -\frac{U_c R_6}{R_5} = -U_c \\ U'_L = -U_L \end{cases} \quad (2)$$

Solving (2), the relationship between output current I_L and input voltage U_i is given by:

$$I_L = \frac{U_i}{R_s} \quad (3)$$

The output current of the circuit is determined solely by the input voltage U_i and the sampling resistor R_s , and is independent of the load. With a 250Ω sampling resistor, the differential VCCS can achieve up to 3 mAp output current at frequencies as high as 200 kHz.

By appropriately configuring the DPDT switch (denoted as MUX3) together with a single-pole/double-throw (SPDT) switch (denoted as MUX2), any two DAEs can be flexibly assigned as an excitation electrode pair. As shown in Fig. 1, under the “skip-1” excitation pattern, the differential VCCS at DAE i sources current into the human body, and the return path passes through DAE ($i-2$) and DAE ($i-1$) before the current sinks back into the same VCCS at DAE i .

D. Local Microcontroller Unit

To reduce wiring complexity between the DAEs and the central control hub, each DAE incorporates an STM32G0 MCU, which locally manages excitation timing, configures AFE parameters, and bridges a single-wire half-duplex Universal Asynchronous Receiver/Transmitter (UART) to a 4-wire Serial Peripheral Interface (SPI) interface.

All control signal lines for MUX1~MUX3 are driven by the MCU. The MCU executes control sequences stored in internal Flash memory according to a predefined excitation–measurement pattern, enabling or disabling signal paths in real time during EIT system operation. Each AD5941 within the DAE interfaces with the local MCU via SPI. The MCU receives configuration commands from the central control hub over a half-duplex UART, programs the AD5941 accordingly, and triggers synchronized current injection or voltage acquisition on the rising edge of the global SYNC signal. Demodulated voltage data are then read by the MCU and relayed to the central control hub. Each MCU also dynamically adjusts the measurement module gain based on the amplitude of the demodulated voltage signal, ensuring consistent signal levels across DAEs at different spatial positions.

III. DAE-EIT SYSTEM OPERATION AND CONTROL LOGIC

Fig. 4 depicts the communication architecture of a DAE-EIT system comprising N proposed DAEs. A graphical user interface (GUI) running on the host computer allows the user to interactively control the system and view EIT reconstruction images in real time. The central control hub configures each DAE, ensures that all DAEs operate synchronously according to the predefined excitation–measurement pattern, and transmits the acquired voltage data back to the host computer.

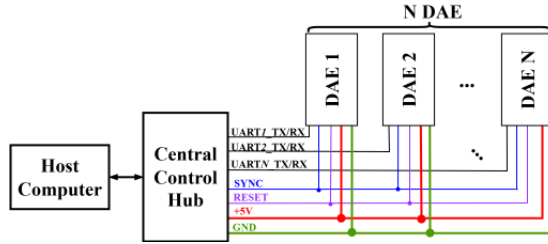


Fig. 4. Interconnection of Host Computer, Central Control Hub, and DAEs.

A. Workflow of each DAE

A typical EIT measurement cycle consists of multiple sequential rounds, each involving current injection through one selected electrode pair while the remaining electrodes simultaneously acquire boundary voltages.

To enable automatic gain adjustment, each measurement round comprises two sequential phases: pre-measurement and measurement. During the pre-measurement phase, the local MCU within each DAE sets the measurement module's gain to unity and performs a 128-point DFT for rapid demodulation to estimate the boundary voltage amplitude. In the measurement phase, the MCU determines the optimal gain by comparing the pre-measured voltage amplitude with sixteen predefined voltage intervals, defined by the ADC's linear input range and available gain coefficients. It then configures a 512-point DFT to acquire the accurate boundary voltage amplitude and transmits it to the central control hub. The 128-point DFT shortens the computation time by about 0.95 ms compared with

the 512-point DFT, at the cost of a 6 dB SNR loss, thus being used only for pre-measurement.

Consequently, each DAE takes 2.93 ms to complete a single round of voltage acquisition. In a 16-electrode EIT system based on the proposed DAE, all DAEs operate in parallel, resulting in a measurement frame rate of 21.33 frames per second (fps).

B. Communication Protocol

As the number of DAEs in the DAE-EIT system increases, low-complexity wiring becomes essential. By integrating an MCU within each DAE, a scalable two-wire communication interface is established between all DAEs and the central control hub, as illustrated in Fig. 4.

Each DAE communicates with the central control hub via a half-duplex UART interface, using UART_{*i*}_TX/RX signal lines (i indicates the DAE index). A global synchronization pulse is generated by the central control hub and distributed to all DAEs via the shared SYNC signal line. All DAEs are triggered to initiate a measurement round on the rising edge of SYNC. Consequently, for a DAE-EIT system comprising N proposed DAEs, only $N + 1$ signal lines are required for data communication and synchronization. Moreover, three additional lines are shared across all DAEs for power and control: the power supply (+5V), common ground (GND), and the reset signal (RESET).

C. FPGA Board of the Central Control Hub

The central control hub consists of a core board and a baseboard, enabling flexible adaptation to EIT systems with varying numbers of electrodes. The core board is built around the Xilinx Zynq-7020 series System-on-Chip (SoC). The ARM processor subsystem (PS) manages host communication and system-level control, whereas the FPGA fabric (PL) implements hardware controllers for multiple DAEs. The baseboard provides peripheral interfaces and connectors for multiple DAEs, along with system-wide power distribution.

IV. RESULTS AND DISCUSSION

To evaluate the performance of the proposed DAE, a DAE-EIT system prototype comprising 16 DAEs connected to a saline tank was built, as shown in Fig. 5. Each DAE has a compact form factor of approximately 7.11 cm × 2.54 cm and a weight of about 7.2 g.

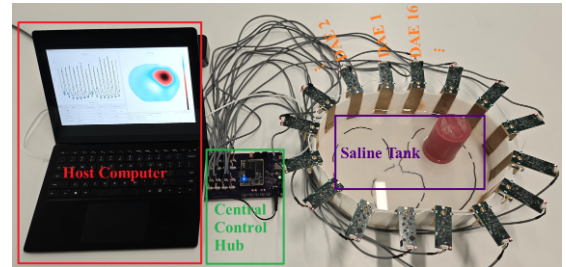


Fig. 5. The prototype of the proposed DAE-EIT system.

A. Signal-to-Noise Ratio

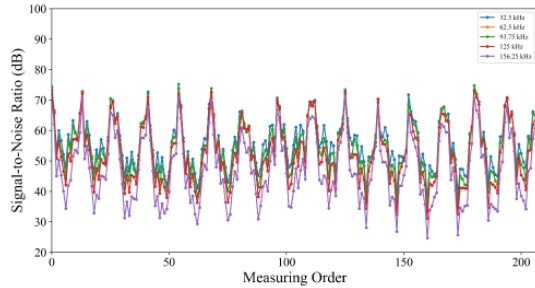
The SNR is a measure of the strength of the desired signal relative to the background noise and can be given by:

$$\text{SNR(dB)} = 10 \times \log_{10} \frac{\sum_{i=1}^N x_i^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \quad (4)$$

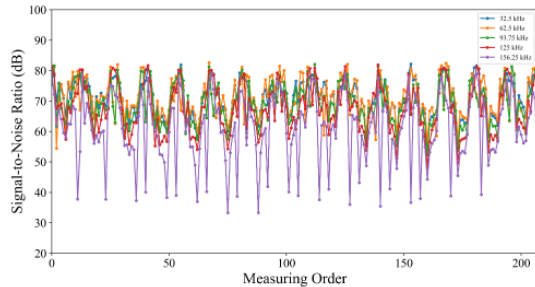
where x_i denotes the i -th measured boundary voltage data and $\bar{x}$ represents the mean value over N measurements. $N = 5000$ is used in the test.

The test object is a mixed solution of saline and distilled water contained in the saline tank, resulting in a background conductivity of 0.75 mS/cm. The excitation signal is a sinusoidal current with an amplitude of 1 mA at five different frequencies, using an adjacent excitation and adjacent measurement pattern. The SNR test results, with and without the proposed adaptive gain control, are shown in Fig. 6.

With the proposed adaptive gain method, both the average SNR of the system and the SNR consistency across different channels are improved. Taking 125 kHz as an example, the system's average SNR increased from 52.60 dB to 67.87 dB, while the maximum SNR difference among the DAEs decreased from 46.59 dB to 33.83 dB. Moreover, the system inherently exhibits better noise suppression and inter-channel consistency at mid-to-low frequencies compared with higher frequencies. At 62.5 kHz, with the proposed adaptive gain method, the system exhibited its optimal performance, with an average SNR of 70.87 dB and a maximum inter-DAE SNR difference of 28.18 dB.



(a) The adaptive gain control is not applied.



(b) The adaptive gain control is applied.

Fig. 6. SNR test results at five frequencies.

B. Imaging Results

The imaging performance of the proposed DAE-EIT system is assessed with four different conductivity distribution models

(denoted as Case 1~Case 4). In all cases, the saline tank is filled with a saline solution having the same conductivity as that used in the SNR tests, and a sinusoidal current of 1 mA at 62.5 kHz is applied for excitation, with both current stimulation and voltage measurement based on adjacent patterns. The high-conductivity objects were made of aluminum with a diameter of 5.8 cm, while the low-conductivity objects were made of nylon with a diameter of 4.1 cm and 6.1 cm. The imaging results obtained with the Newton's One-Step Error Reconstructor algorithm are shown in Fig. 7 [10].

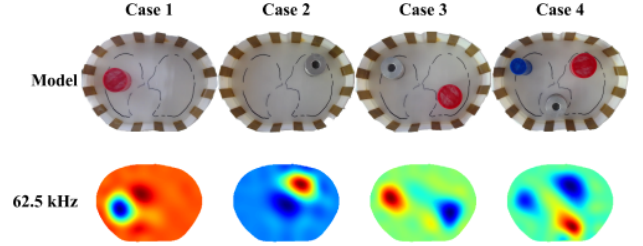


Fig. 7. Image reconstruction results under different models at 62.5 kHz.

The imaging results demonstrate that the reconstructed conductivity distributions accurately capture the spatial locations, morphological contours, and relative conductivity contrasts of both low- and high-conductivity objects within the tank. Image quality could be further improved with alternative EIT reconstruction algorithms.

V. CONCLUSIONS

In this paper, an AD5941-enabled DAE for EIT is proposed. Each DAE integrates both the excitation and measurement modules at the electrode side to eliminate the parasitic effects of long lead wires. To improve the system's noise suppression capability and inter-channel consistency, adaptive gain control is introduced. Based on this design, a 16-electrode DAE-EIT system has been developed and tested in a saline tank, achieving an average SNR above 70 dB at 62.5 kHz with 21.33 fps. Future work will focus on reducing the size of the DAE and conducting imaging experiments using phantoms based on biological tissue.

REFERENCES

- [1] D. Khodadad, S. Nordebo, B. Müller, *et al.*, "Optimized breath detection algorithm in electrical impedance tomography," *Physiological Measurement*, vol. 39, no. 9, p. 094001, 2018.
- [2] J.-B. Li, C. Tang, M. Dai, *et al.*, "A new head phantom with realistic shape and spatially varying skull resistivity distribution," *IEEE Transactions on Biomedical Engineering*, vol. 61, no. 2, pp. 254–263, 2013.
- [3] D. Yang, G. Huang, B. Xu, X. Wang, Z. Wang, and Z. Wei, "A DSP-based EIT system with adaptive boundary voltage acquisition," *IEEE Sensors Journal*, vol. 22, no. 6, pp. 5743–5754, 2022.
- [4] N. Sun, D. Yang, B. Xu, and S. Li, "Development of a miniaturized, low-cost electrical impedance tomography system," *IEEE Transactions on Instrumentation and Measurement*, 2024.
- [5] M. H. Lee, G. Y. Jang, Y. E. Kim, *et al.*, "Portable multi-parameter electrical impedance tomography for sleep apnea and hypoventilation monitoring: Feasibility study," *Physiological Measurement*, vol. 39, no. 12, p. 124004, 2018.

- [6] Y. Wu, D. Jiang, A. Bardill, R. Bayford, and A. Demosthenous, "A 122 fps, 1 MHz bandwidth multi-frequency wearable EIT belt featuring novel active electrode architecture for neonatal thorax vital sign monitoring," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 13, no. 5, pp. 927–937, 2019.
- [7] C. Wang, W. Lu, J. Huang, *et al.*, "Flexi-EIT: A flexible and reconfigurable active electrode electrical impedance tomography system," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 18, no. 1, pp. 89–99, 2023.
- [8] R. B. Santos, A. L. Santos, A. C. M. Cavaleiro, *et al.*, "Electrode module for EIT with a robust howland current source," *IFAC Journal of Systems and Control*, vol. 28, p. 100265, 2024.
- [9] A. S. Ross, G. J. Saulnier, J. C. Newell, and D. Isaacson, "Current source design for electrical impedance tomography," *Physiological Measurement*, vol. 24, no. 2, p. 509, 2003.
- [10] M. Cheney, D. Isaacson, J. C. Newell, S. Simske, and J. Goble, "NOSER: An algorithm for solving the inverse conductivity problem," *International Journal of Imaging Systems and Technology*, vol. 2, no. 2, pp. 66–75, 1990.